

Recall: Let $f_1, f_2, \dots, f_N$ be maps on $D \subseteq \mathbb{R}^n$ such that

$$b_i \|x-y\| \leq \|f_i(x) - f_i(y)\| \leq c_i \|x-y\| \quad \forall x, y \in D,$$

where $0 < b_i \leq c_i < 1 \quad \forall i = 1, 2, \dots, N$.

Suppose $f_i(K) \cap f_j(K) = \emptyset$ for $i \neq j$, where K is the attractor of the IFS.

Then $s_0 \leq \dim_H K \leq t_0$, where

$$\sum_{i=1}^N b_i^{s_0} = 1 \quad \text{and} \quad \sum_{i=1}^N c_i^{t_0} = 1.$$

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Note that $K = \bigcup_{i=1}^N f_i(K)$ gives

$$K = \bigcup_{i_1, i_2, \dots, i_m} f_{i_1} \circ f_{i_2} \circ \dots \circ f_{i_m}(K)$$

$\therefore K$ is the attractor of the IFS given by the maps $\{f_\omega : \omega \in \{1, 2, \dots, N\}^m\}$ for any $m \in \mathbb{N}$.

(If $\omega = (i_1, i_2, \dots, i_m)$, then $f_\omega := f_{i_1} \circ f_{i_2} \circ \dots \circ f_{i_m}$)

Then we find $0 < b_\omega \leq c_\omega < 1$, such that

$$b_\omega \|x-y\| \leq \|f_\omega(x) - f_\omega(y)\| \leq c_\omega \|x-y\| \quad \forall x, y \in D$$

Then $s'_0 \leq \dim_H K \leq t'_0$, where

$$\sum_{\omega \in \mathbb{I}^m} b_\omega^{s'_0} = 1 \quad \text{and} \quad \sum_{\omega \in \mathbb{I}^m} c_\omega^{t'_0} = 1.$$

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Remark: Using large enough m , we can find better lower and upper bounds for $\dim_H K$.

Example: (Continued fractions)

Let $x \in (0,1)$. Then x can be written in form $x = \frac{1}{n_1 + \frac{1}{n_2 + \frac{1}{n_3 + \dots}}}$, where each $n_i \in \mathbb{N}$.

Consider the set $E = \{x \in (0,1): \text{the continued fraction expansion of } x \text{ has digits either 1 or 2}\}$

Q: What is $\dim_H E$?

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Let $f_1, f_2: [0,1] \rightarrow [0,1]$ be given by

$$f_1(x) = \frac{1}{1+x}$$

$$f_2(x) = \frac{1}{2+x}$$

Then $E = f_1(E) \cup f_2(E)$

Also, $f_1(E) \cap f_2(E) = \emptyset$

Note that f_1 is not a strict contraction on $[0,1]$.

What is $\inf E$?

$$\inf E = \frac{1}{2 + \frac{1}{1 + \frac{1}{2 + \dots}}} = \alpha$$

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$$\alpha = \frac{1}{2 + \frac{1}{1 + \frac{1}{2 + \dots}}} = \frac{1}{2 + \frac{1}{1 + \alpha}} = \frac{1 + \alpha}{3 + 2\alpha}$$

$$\Rightarrow 2\alpha^2 + 3\alpha = 1 + \alpha$$

$$\Rightarrow 2\alpha^2 + 2\alpha - 1 = 0$$

$$\Rightarrow \alpha = \frac{-2 + 2\sqrt{3}}{4} = \frac{\sqrt{3} - 1}{2}$$

$$\beta = \sup E = \frac{1}{1 + \frac{1}{2 + \frac{1}{1 + \dots}}} = \frac{1}{1 + \frac{1}{2 + \beta}} = \frac{2 + \beta}{3 + \beta}$$

$$\Rightarrow \beta^2 + 3\beta = 2 + \beta$$

$$\Rightarrow \beta^2 + 2\beta - 2 = 0 \Rightarrow \beta = \frac{-2 + 2\sqrt{3}}{2} = \sqrt{3} - 1$$

$\therefore$ We can take $f_i : \left[\frac{\sqrt{3}-1}{2}, \sqrt{3}-1 \right] \rightarrow D$
 $f_i(x) = \frac{1}{i+x}, \quad i=1, 2$

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$\therefore E$ is the unique attractor of the IFS $\{f_1, f_2\}$

$$f_1(x) = \frac{1}{1+x}$$

$$\Rightarrow |f_1'(x)| = \frac{1}{(1+x)^2}$$

$$\Rightarrow \max_{x \in D} f_1'(x) = \frac{1}{\left(1 + \frac{\sqrt{3}-1}{2}\right)^2} = \frac{4}{(\sqrt{3}+1)^2}$$

$$\& \min_{x \in D} f_1'(x) = \frac{1}{(1 + \sqrt{3}-1)^2} = \frac{1}{3}$$

$$\therefore \text{For } f_1, \quad c_1 = \frac{4}{(\sqrt{3}+1)^2}, \quad b_1 = \frac{1}{3}$$

Similarly, we can find c_2 & b_2 for f_2

$$\frac{1}{\left(2 + \frac{\sqrt{3}-1}{2}\right)^2} \quad \frac{1}{(2 + \sqrt{3}-1)^2}$$

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Then we find s_0 s.t. $b_1^{s_0} + b_2^{s_0} = 1$ } (Exercise)
 and t_0 s.t. $c_1^{t_0} + c_2^{t_0} = 1$

This gives $s_0 \leq \dim_H E \leq t_0$.

How to improve these bounds?

Take the IFS = $\{f_1 \circ f_1, f_1 \circ f_2, f_2 \circ f_1, f_2 \circ f_2\}$

Then E is also the attractor of this IFS.

Then we find b_i 's & c_i 's for these maps
 and use them to find s_0 & t_0 .

$$f_1 \circ f_1(x) = f_1\left(\frac{1}{1+x}\right) = \frac{1}{1 + \frac{1}{1+x}} = \frac{1+x}{2+x}$$

$$(f_1 \circ f_1)'(x) = \frac{1 \cdot (2+x) - 1 \cdot (1+x)}{(2+x)^2} = \frac{1}{(2+x)^2}$$

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$$\therefore \max_{x \in D} |(f_1 \circ f_1)'(x)| = \frac{1}{\left(2 + \frac{\sqrt{5}-1}{2}\right)^2} = \frac{4}{(3+\sqrt{5})^2}$$

$$\min_{x \in D} |(f_1 \circ f_1)'(x)| = \frac{1}{(2 + \sqrt{3}-1)^2} = \frac{1}{(\sqrt{5}+1)^2}$$

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